

Characterization of EEG signals using wavelet transform for motor imagination tasks in BCI systems

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Abstract—Brain Computer Interface (BCI) covers an area of special interest, mainly due to the research being conducted to control external devices via thought commands, generating solutions in both motor disability and speech. These applications require the use of signal processing in real time, to be used in devices that help people with this type of disabilities. This paper presents a methodology for feature extraction of electroencephalographic (EEG) signals in motor imagery task of both left and right hand, using the public database BCI Competition 2003. It has been used the wavelet transform for the signal decomposition in the spectral bands of interest (known as brain rhythms). The brain rhythms characterization was conducted through relative energy, variance and standard deviation of the wavelet coefficients. In addition, we conducted the relevance analysis through the fuzzy entropy algorithm, to find the most important features within the training set. We obtained a classification accuracy of up to 98.44% using K-NN and SVM algorithms. The classification results allow inferring that the methodology is appropriate for the recognition of imagination movements in people with motor disabilities and could generate solutions in applications of BCI systems.

Keywords—Discrete Wavelet Transform; BCI; EEG; fuzzy entropy; feature extraction.

I. INTRODUCTION

Left and right hand motor imagery can modify neuronal activity in major sensorimotor areas, leading to changes in the rhythms mu (μ) and beta (β). BCI systems require an effective method of online processing to classify these EEG signals in order to build a system that allows physically disabled patients communicate with their environment [1-2].

Several techniques have been used to extract features for this purpose, including autoregressive models [3,4], common spatial patterns [5], and wavelet transform [6,7]. This paper presents a methodology for feature extraction of electroencephalographic (EEG) signals in motor imagery task of both left and right hand. It is used Discrete Wavelet Transform (DWT) to decompose the EEG signal in their bands of interest. Then brain rhythms are characterized with statistical functions such as relative energy, variance and standard deviation of the wavelet coefficients. Additionally employs fuzzy entropy algorithm for relevance analysis that determines the most important features in the training set. Finally, SVM (support Vector machines) and K-NN (K-nearest neighbors) are used to classify the reduced feature set.

The application of this method was used to analyze the data set provided by Graz BCI competition II, data set III.

II. METHODS

A. Discrete Wavelet Transform (DWT)

Wavelet Transform has shown promising results in the treatment of EEG signals due to non-stationary characteristics of the signal, besides reflecting the coefficients in the time-frequency domain and multi-resolution decomposing ability.

Specifically, discrete wavelet transform decomposes the signal approximation coefficients (A) and the detail coefficients (D). The approximation coefficient then divides into new approximation coefficients and detail. This process is performed iteratively, producing a set of approximation coefficients and detail coefficients at different levels or scales [8].

Wavelet coefficients can allow a smaller number of features for representing of the original signal. For a signal $x(t)$, the equation of DWT decomposition is given by (1):

$$x(t) = \sum_{k=-\infty}^{k=+\infty} c_{n,k} \phi(2^{-n}t - k) + \sum_{j=1}^n \sum_{k=-\infty}^{k=+\infty} d_{j,k} 2^{-j/2} \Psi(2^{-j}t - k) \quad (1)$$

Where $C_{n,k}$ represents the coefficients to approach the level n while $d_{j,k}$ ($j=1$ to n) represents the detail coefficients at level j , $\Psi(t)$ is the wavelet function and $\phi(t)$ is a function of scale. The mother wavelet selection determines the signal representation. Generally Daubechies, Symlet and Coiflet mother wavelet families are used. The extracted coefficients from wavelet decomposition are too long to be used as features for classification; in this paper we used three statistical functions to characterize the wavelet coefficients in each of the sub-levels of decomposition.

B. Fuzzy Entropy

In the probabilistic approach, the Shannon entropy is a well known measure of uncertainty and is covered extensively in the literature. An extension of the Shannon entropy is the concept of fuzzy entropy where the fuzzy sets used to assist estimation of entropy. The measuring of the fuzzy entropy is quite different from the classical Shannon entropy since fuzzy

entropy contains fuzziness uncertainties (possibilistic), while Shannon entropy contains randomness uncertainties (probabilistic).

The fuzzy entropy, like Shannon entropy satisfies the four axioms of De Luca-Termini [9]. The Shannon entropy is defined as a discrete random variable (x) with probability function $p(x)$ given by (2).

$$H(X) = - \sum_i p(x_i) \log_2 p(x_i) \quad (2)$$

From Shannon entropy, fuzzy joint entropy of the elements of the class i , denoted as $H(f, c_i)$, is defined by (3).

$$H(f, c_i) = -P(f, c_i) \log P(f, c_i) \quad (3)$$

Complete entropy fuzzy along the c -class is given by (4):

$$H(f, C) = \sum_i^c H(f, c_i) \quad (4)$$

The fuzzy equivalent to the joint probability of the training patterns belonging to the class i is given as:

$$P(f, c_i) = \frac{\sum_{k \in A_i} \mu_{ik}}{NP} \quad (5)$$

Where A_i is the set of indices of the training patterns belonging to class i , NP is the total number of patterns, and μ_{ik} is the k -th fuzzy membership value of belonging to class i .

Constructing the fuzzy membership values for each of the features obtained, the algorithm is performed by fuzzy c -means; This method, often used in pattern recognition, are based on minimizing the objective function expressed in (6), where m is a real number greater than 1, x_k is the k -th sample, c_i are the centers of the clusters, $\|\cdot\|$ is the Euclidean norm expressing the similarity between the measured data and the centroid, the membership value of x_k in the cluster i , is expressed in (7).

$$J_m(P, V) = \sum_{k=1}^N \sum_{i=1}^c \mu_{ik}^m \|x_k - c_i\|^2 \quad 1 \leq m \leq \infty \quad (6)$$

$$\mu_{ik} = \frac{1}{\sum_{j=1}^c \left(\frac{\|x_k - c_i\|}{\|x_k - c_j\|} \right)^{\frac{2}{m-1}}} \quad (7)$$

III. EXPERIMENTAL FRAMEWORK

We used data collected from online BCI experiments, and analyzed as described in this section.

A. Dataset

The data set was provided by the Department of Medical Informatics, Institute of Biomedical Engineering, Graz University of Technology [10].

This dataset was recorded from a normal subject (female, 25y) during a feedback session. The subject sat in a relaxing chair with armrests. The task was to control a feedback bar by means of imagery left or right hand movements. The order of left and right cues was random. The positions of the electrodes and the distribution system are shown in Fig. 1.

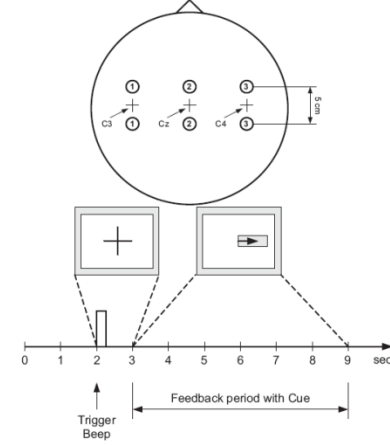


Fig. 1. Electrode positions (up) and timing (down) scheme.

The experiment consists of 7 runs with 40 trials each. All runs were conducted on the same day with several minutes break in between. Given are 280 trials of 9s length. The first 2s was quite, at $t=2s$ an acoustic stimulus indicates the beginning of the trial, the trigger channel (#4) went from low to high, and a cross “+” was displayed for 1s; then at $t=3s$, an arrow (left or right) was displayed as cue. At the same time the subject was asked to move a bar into the direction of the cue. The recording was made using a G.tec amplifier and a Ag/AgCl electrodes. Three bipolar EEG channels (anterior ‘+’, posterior ‘-’) were measured over C3, Cz and C4. The EEG was sampled with 128Hz, it was filtered between 0.5 and 30Hz.

The trials for training and testing were randomly selected. This should prevent any systematic effect due to the feedback.

B. Experiments

Because channel Cz showed its independence of the motor imagery, only channel C3 and C4 were used for feature extraction [11]. Firstly, we calculated the average power spectral density (PSD) of motor imagination tasks for the respective channels C3 and C4 in order to effectively provide a reference in the wavelet decomposition step.

The power spectrum at C3 and C4 of the training set is shown in Fig. 2. This indicates that the power spectrum is mainly distributed in 8-11 Hz and 19-22 Hz ranges, typically associated with mu (μ) and beta (β) rhythms, respectively.

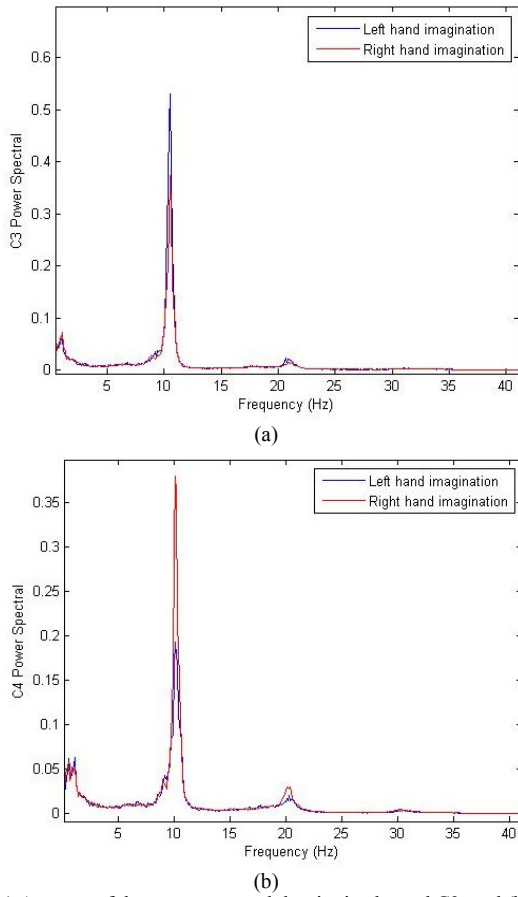


Fig. 2. (a) Average of the power spectral density in channel C3, and (b) average power spectral density in channel C4.

According to EEG motor imagery signals and power spectrum in channels C3 and C4, were chosen five levels of decomposition and mother wavelet Daubechies order 10. Table 1 shows frequency bands obtained for each level of wavelet decomposition with 128Hz sampling frequency..

TABLE I. WAVELET DECOMPOSITION LEVELS.

Decomposition level	Frequency
D1	64 - 128 Hz
D2	32 - 64 Hz
D3	16 - 32 Hz
D4	8 - 16 Hz
D5	4 - 8 Hz
A5	0 - 4 Hz

The extracted wavelet coefficients show the distribution of motor imagination signal in time and frequency. In order to characterize brain rhythms and represent the distribution of the frequency and the number of changes therein, three statistical functions are calculated on the set of wavelet coefficients for each sub-frequency band, namely:

- Standard Deviation
- Relative Energy Wavelet
- Variance

Thus, thirty-six features of wavelet coefficients are obtained for the two channels.

Using the fuzzy c-means algorithm, we construct the fuzzy membership values for each of the features obtained, with $c_k=2$. Each of these reflects the degree of membership of the sample for each of the two classes.

Combining μ_{ik} membership values obtained by the equations (3), (4) and (5), fuzzy entropy was calculated on the specific feature space.

Applying Fuzzy Entropy algorithm to the feature set, the dimensionality of the feature set is reduced, based on the fact that a value of high entropy contributes little to the deviation between classes and low entropy values more informative exhibit characteristics [12]. Thus, the ten characteristics with less fuzzy entropy value were trained and classified by cross-validation (10 fold cross-validation) for k-nn classifiers and support vector machines (SVM).

Fig. 3 shows in detail the structure and proposed method for characterizing EEG signals in motor imagination tasks.

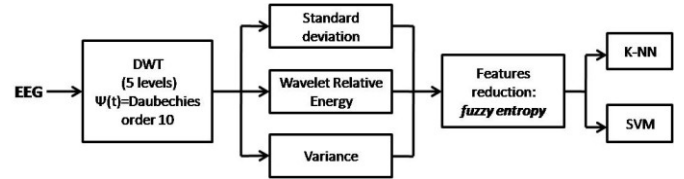


Fig. 3. Scheme of the proposed method to characterize EEG signals in motor imagination tasks.

C. Results and discussion

Classification results of the features extracted by the proposed method are shown in Table (2) for k-nn classifier ($k=3$) and SVM classifier with linear kernel function to map the training points to a higher-dimensional space.

The results show that SVM produces the best performance, with an accuracy of 98.44%, but the K-NN classifier also proves effective for pattern classification in motor imagery tasks of both left and right hand.

Furthermore, in table (2) can also be inferred that the proportion of correctly identified classes is higher than 97.39% for the best of the classifiers, evidenced parameters from the sensitivity and specificity in the classification algorithm.

TABLE II. CLASSIFICATION RESULTS OF THE PROPOSED METHOD USING SVM AND K-NN CLASSIFIER.

Measures (%)	Classifiers	
	SVM	K-NN
Accuracy	98.44 ± 1.46	93.05 ± 2.94
Sensitivity	97.39 ± 2.31	95.70 ± 3.33
Specificity	99.30 ± 1.61	90.40 ± 4.37

Other patterns extraction and classification techniques applied on the data set used in this work show inferior results, as in [13] and [14] of 90%, [15] with better ranking results on average of 86.51%, and in [11] with misclassification of 3.94%.

IV. CONCLUSIONS AND FUTURE WORK

This paper proposed a method for feature extraction of EEG signal for imagination of right and left hand movement. The pattern classification techniques described in this paper enable the automated development of BCI systems based on thought pattern recognition of EEG signals.

Future work includes optimize the algorithm to improve classification accuracy, use this method in other BCI databases, and implement a BCI system for people with motor disabilities.

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